CIE AS PHYSICS (9702) • QUESTION PAPER

6.1 Stress & Strain

Syllabus Section: 6. Deformation of Solids **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (99 Marks)

Question 1 Easy [11 marks]

Part (a) [3 marks]

Forces can change the motion of a body, or cause deformation.

State

(i) the definition of *deformation*.

[1]

(ii) the name of the force which stretches an object.

[1]

(iii) the name of the force which squashes an object.

[1]

Part (b) [2 marks]

Hooke's Law can be applied to objects which are stretched or compressed.

State the equation for Hooke's Law, defining any terms used.

Part (c) [3 marks]

A spring is suspended from a stand as shown in Fig. 1.1 and has a load of 5.0 N applied to it. The spring extends by 4.2 cm.

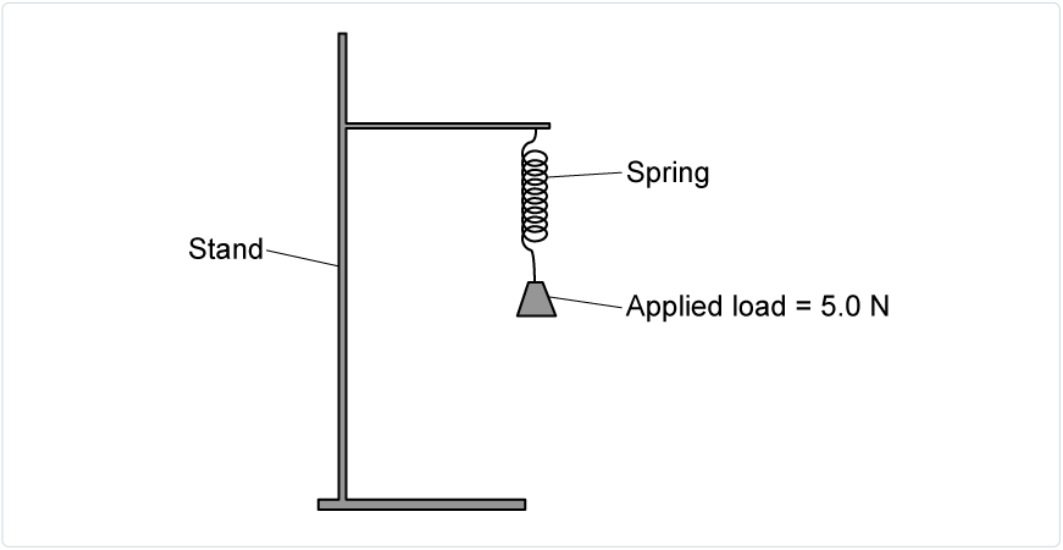


Fig. 1.1

Calculate the spring constant, expressing your answer in S.I. units.

Part (d) [3 marks]

Another, thicker spring is attached to a base so that it stands upright, as shown in Fig. 1.2.

A force of 6.3 N pushes down on the spring which compresses it by 2.4 cm.

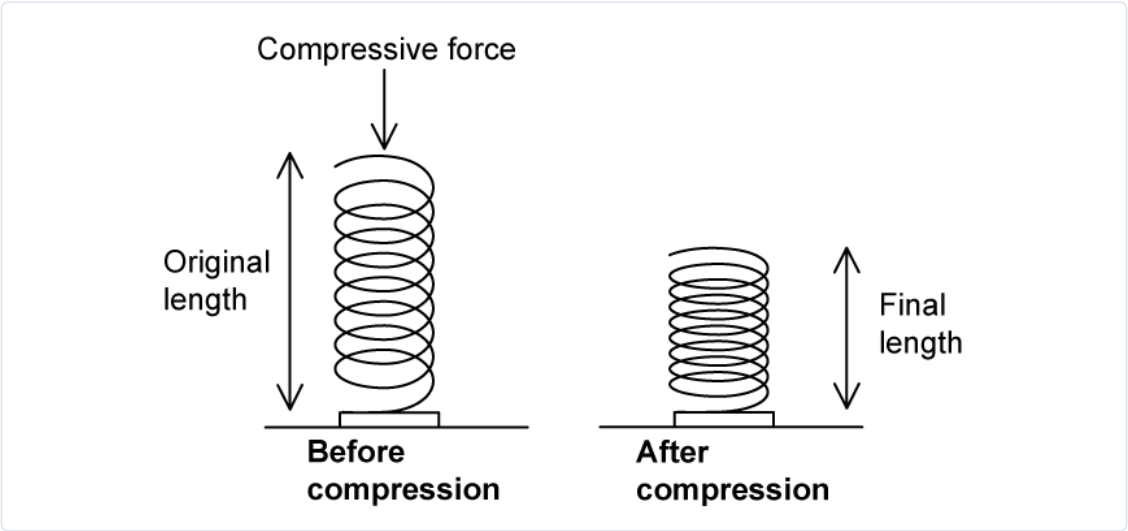


Fig. 1.2.

Calculate the spring constant, expressing your answer in S.I. units.

Question 2 Easy [15 marks]

Part (a) [3 marks]

Define

(i) tensile stress

[1]

(ii) tensile strain

[1]

(iii) the Young modulus.

[1]

Part (b) [4 marks]

Fig. 1.1 shows stress-strain graphs for two different metal wires X and Y.

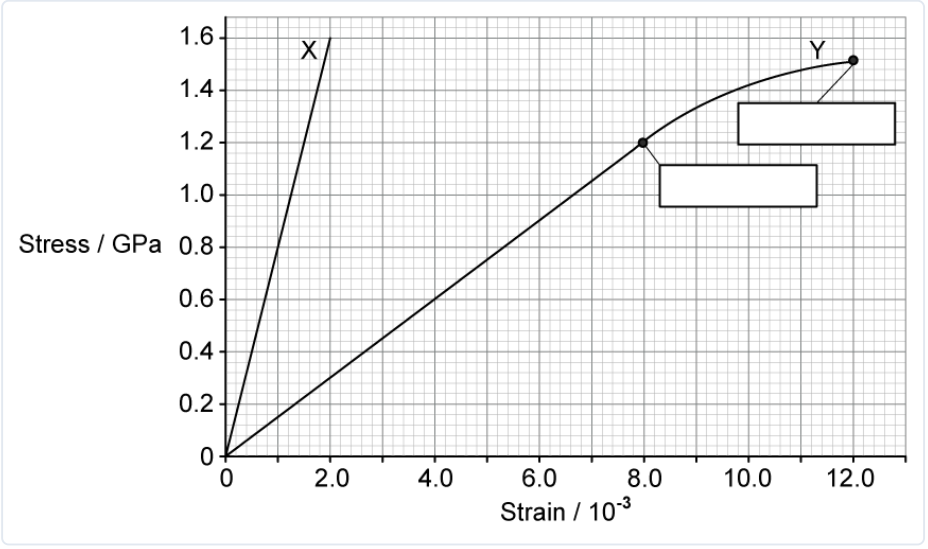


Fig. 1.1

Fill in the missing labels on Fig. 1.1 and state the definitions of these terms.

Part (c) [2 marks]

State and explain which wire, X or Y, has the greater Young modulus.

Part (d) [6 marks]

Use the graph in Fig. 1.1 to determine

(i) the Young modulus of wire X and state the unit.

[3]

(ii) the extension of wire Y when the stress is 1.2 GPa, and state the unit.

original length of wire Y = 2.0 m

[3]

Question 3 Easy [11 marks]

Part (a) [5 marks]

A student is carrying out an experiment to determine the Young modulus of a wire. The student sets up the apparatus as shown in Fig. 1.1.

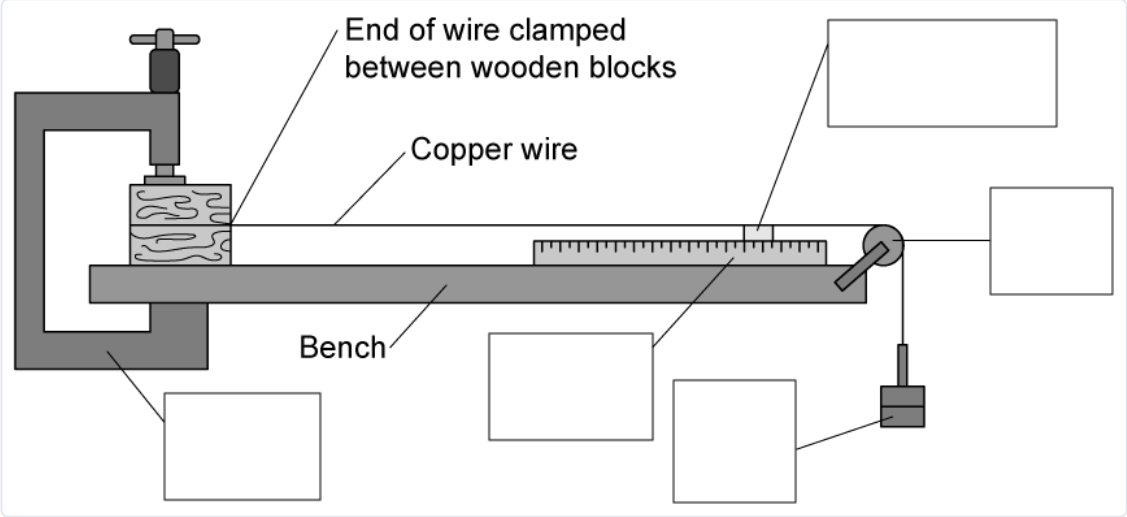


Fig. 1.1

Fill in the missing labels for the apparatus in Fig. 1.1.

Part (b) [2 marks]

State and explain a safety consideration for this experiment.

Part (c) [2 marks]

On the axis of Fig. 1.2 below, draw a graph to show the expected variation of strain with stress for this experiment.

Assume that the wire does not stretch beyond the limit of proportionality.

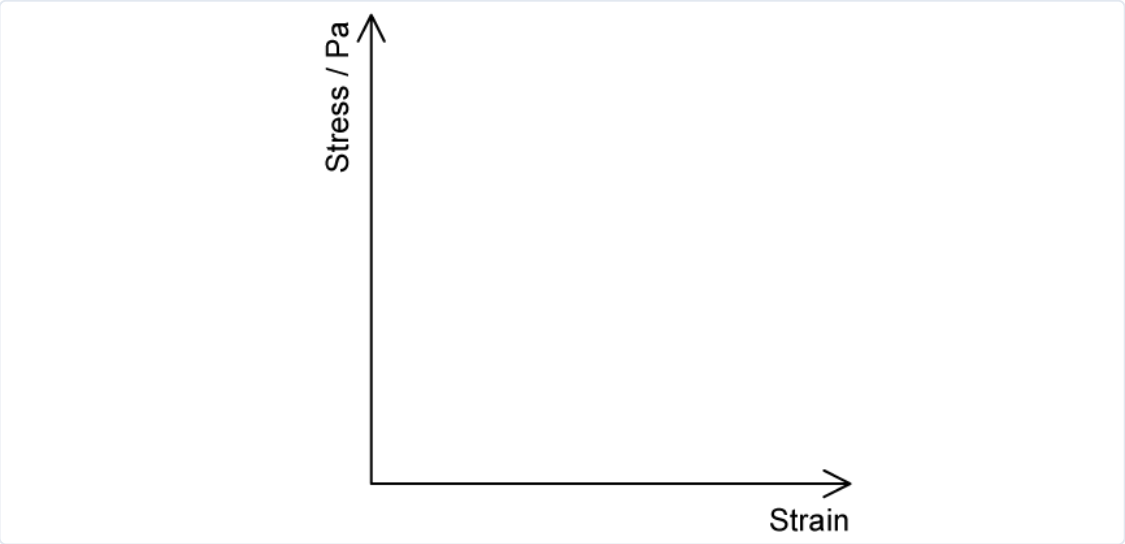


Fig. 1.2

Part (d) [2 marks]

Explain how the student could use the graph in Fig. 1.2 to determine the Young modulus of the wire.

Question 4 Medium [12 marks]

Part (a) [2 marks]

(i) Define force.

[1]

(ii) State what is meant by work done.

[1]

Part (b) [10 marks]

A block of mass 0.40 kg slides in a straight line with a constant speed of 0.30 m s^{-1} along a horizontal surface, as shown in Fig. 3.1.

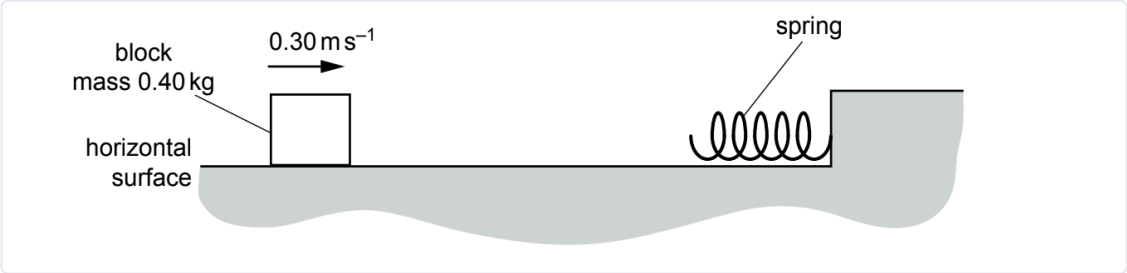


Fig. 3.1

Assume that there are no resistive forces opposing the motion of the block.

The block hits a spring and decelerates. The speed of the block becomes zero when the compression of the spring is 8.0 cm.

(i) Calculate the initial kinetic energy of the block.

kinetic energy = ... J [2]

(ii) The variation of the compression x of the spring with the force F applied to the spring is shown in Fig. 3.2.

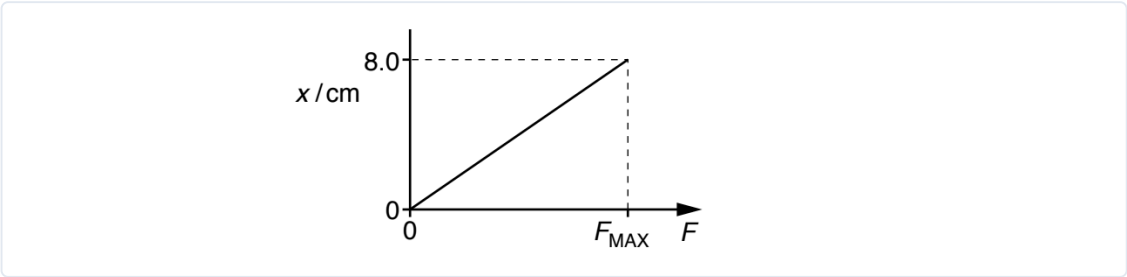


Fig. 3.2

Assume that the elastic potential energy of the spring when its compression is 8.0 cm is equal to the initial kinetic energy of the block.

Use your answer in **(b)(i)** to calculate the maximum force F_{max} exerted on the spring by the block.

F_{max} = ... N [2]

(iii) Calculate the maximum deceleration of the block.

deceleration = ... m s^{-2} [2]

(iv) State and explain whether the block is in equilibrium:

- before it hits the spring

.....

- when its speed becomes zero.

.....

[2]

(v) The block is now replaced by another block of the same mass. Frictional forces affect the motion of this block so that it has a speed of 0.25 m s^{-1} when it makes contact with the spring.

A short time later, the block has a speed of 0.15 m s^{-1} as it loses contact with the spring and moves back along its original path.

Calculate the magnitude of the change in momentum of the block.

change in momentum = ... N s [2]

Question 5 Medium [10 marks]

Part (a) [1 mark]

Define work done by a force.

Part (b) [3 marks]

A heavy suitcase is placed on a weighing machine consisting of a platform mounted on a spring as shown in Fig. 1.1.

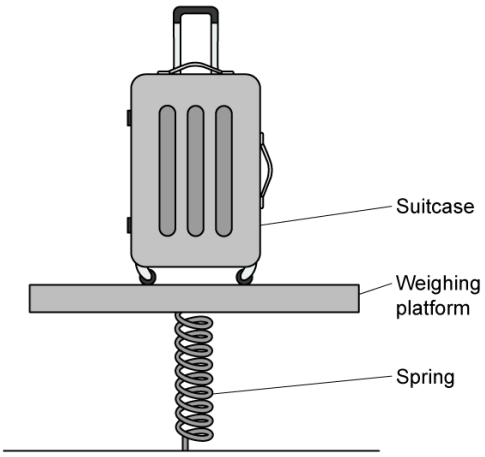


Fig. 1.1

Describe and explain how work is done in this situation.

Part (c) [2 marks]

The diagram in Fig.1.2 shows a suitcase of mass 22.6 kg positioned on the weighing platform.

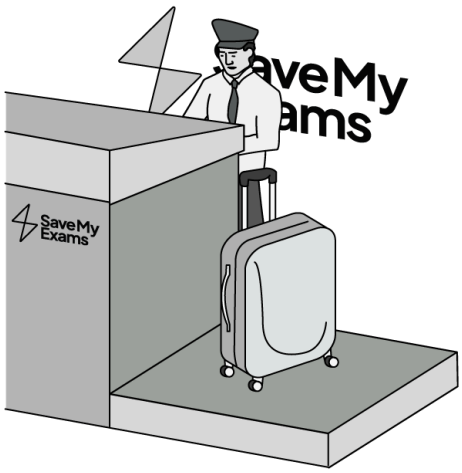


Fig. 1.2

The platform sinks down 3 cm when the suitcase is placed on it, and then returns to its original position when the suitcase is removed.

Calculate the spring constant of the spring.

Part (d) [4 marks]

For the weighing platform described in part (c) sketch a graph to show the known properties of the spring.

Question 6 Medium [9 marks]

Part (a) [2 marks]

Nichrome is a common alloy used to make coins and heating elements in electrical appliances. It consists of 80% by volume of nickel and 20% by volume of chromium.

Determine the mass of nickel and the mass of chromium required to make a wire of nichrome of volume 0.50×10^{-3} m^3.

Density of nickel = 8.9×10^3 kg m^{-3}

Density of chromium = 7.1×10^3 kg m^{-3}

Part (b) [4 marks]

To calculate the breaking stress of the nichrome wire both the cross-sectional area of the wire and the applied load must be known.

For each value, briefly describe an experimental method to determine the value, mentioning any equipment required.

(i) The cross-sectional area of the wire.

[2]

(ii) The applied load.

[2]

Part (c) [3 marks]

The diameter of the nichrome wire is 5.7 mm. The wire breaks under an applied force of 72 kN.

Calculate the breaking stress of nichrome.

Question 7 Hard [13 marks]

Part (a) [3 marks]

A student adds a series of masses to a vertical metal wire with a circular cross–section and measures the extension of the wire produced. This is shown in Fig. 1.1.

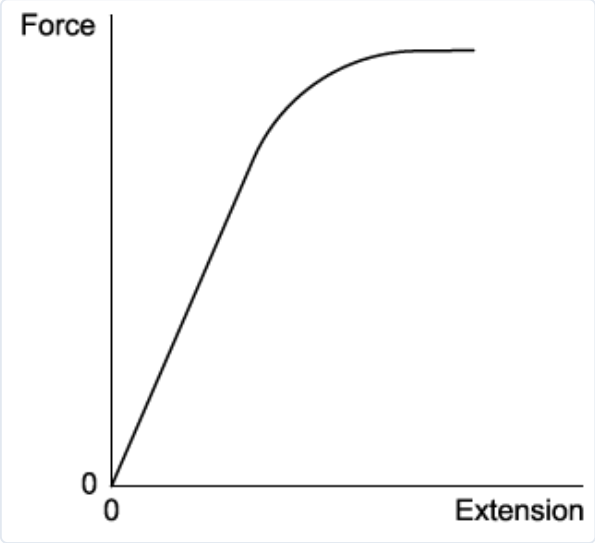


Fig. 1.1

Suggest how the student can determine the Young Modulus of the wire using the graph in Fig. 1.1.

Part (b) [3 marks]

The metal wire is used to make a cable of diameter 4.9 mm. The Young Modulus of the metal in the cable is 154 GPa.

Calculate the force necessary to produce a strain of 0.40 % in the cable.

Part (c) [3 marks]

The cable is used in a crane to lift a mass of *m* kg which is a third of the maximum possible mass it can lift.

Determine the maximum acceleration of the crane when it is lifting an object of mass *m* kg if the strain in the cable does not exceed 0.40%.

Part (d) [4 marks]

Two different cables with spring constants k_1 and k_2 are connected in series.

Starting from Hooke's law, show that the combined spring constant, *K* of the two springs is given by the equation $K = \frac{k_1 k_2}{k_1 + k_2}$.

Question 8 Hard [18 marks]

Part (a) [6 marks]

As part of a quality check, a manufacturer of dog leads selects a sample of the wires used for the leads to a tensile test.

The sample of wires used is 3.0 m long and is of constant circular cross–section of diameter 0.92 mm. Hooke's law is obeyed up to the point when the wire has been extended by 20 mm at a tensile stress of 4.6×10^8 Pa.

The maximum load the wire can support before breaking is 350 N at an extension of 12 cm.

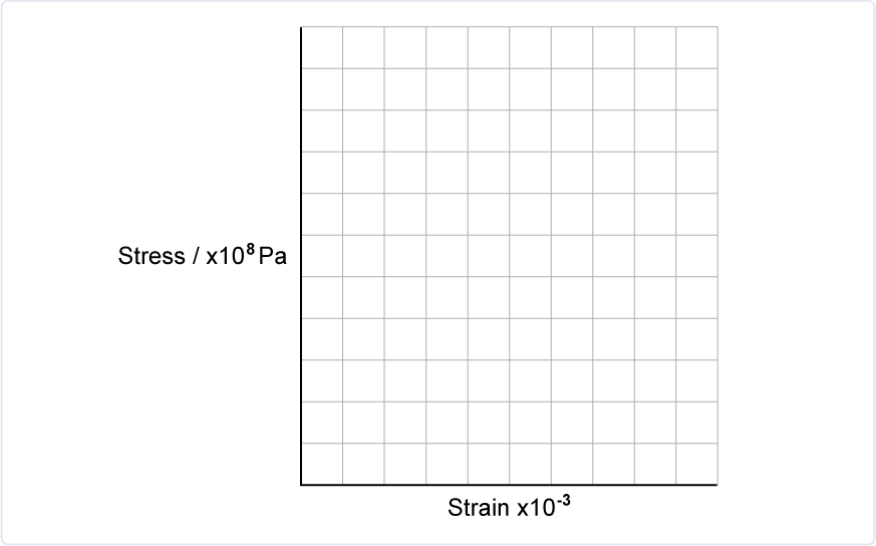


Fig. 1.1

Sketch a graph on Fig. 1.1 to show how you expect the tensile stress to vary with strain and mark the values of stress with corresponding strain at the limit of Hooke's law and the breaking point.

Part (b) [4 marks]

A different 3.0 m dog lead sample has a larger diameter. It is made of the same material and is subject to the same force.

Compare the advantages and disadvantages of this dog lead compared to the first.

Part (c) **[4 marks]**

Calculate the force needed to extend a piece of 3.0 m leather dog leash with a radius of 2.3 mm to change its surface area by 4.1×10^{-3} m^2. Assume that the cross–sectional area is kept constant throughout the extension.

Young Modulus of leather = 0.05 GPa.

Part (d) **[4 marks]**

To prevent the walls of old buildings collapsing a metal rod is often used to tie opposite walls together, as shown in Fig. 1.2 below.

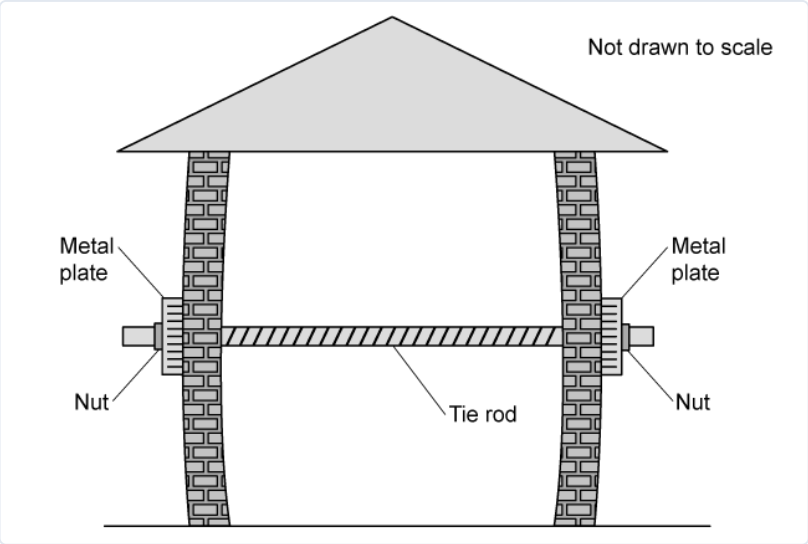


Fig. 1.2

In one case a steel tie rod of diameter 15 mm is used. When the nuts are tightened, the rod extends by 6 %. The Young Modulus of steel is 2.5×10^{11} Pa.

Calculate the force exerted on the walls by the rod.